

Простые сети в GNS3

Лабораторная работа №5. Анализ трафика

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Цели и задачи работы

Изучение принципов построения простейших локальных сетей в среде GNS3 и анализ сетевого трафика с использованием Wireshark.

Моделирование сети с коммутатором

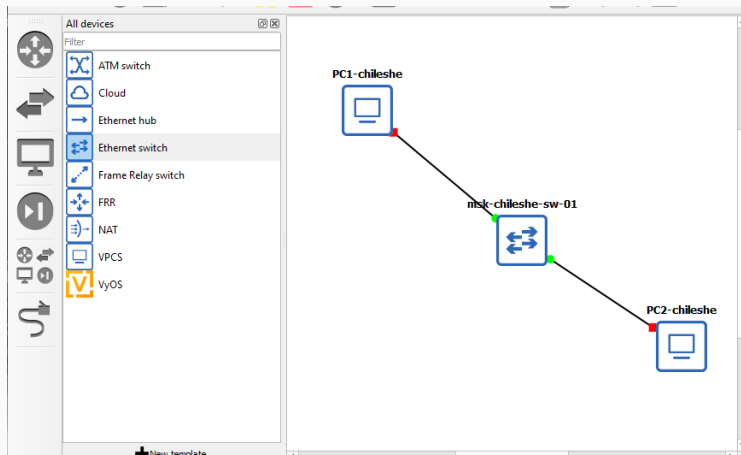
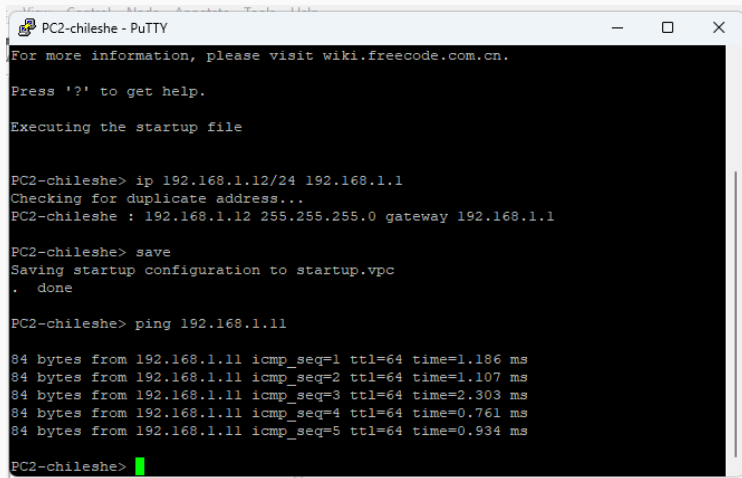


Рис. 1: Топология простейшей сети в GNS3



```
PC2-chileshe - PuTTY
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC2-chileshe> ip 192.168.1.12/24 192.168.1.1
Checking for duplicate address...
PC2-chileshe : 192.168.1.12 255.255.255.0 gateway 192.168.1.1

PC2-chileshe> save
Saving startup configuration to startup.vpc
. done

PC2-chileshe> ping 192.168.1.11

84 bytes from 192.168.1.11 icmp_seq=1 ttl=64 time=1.186 ms
84 bytes from 192.168.1.11 icmp_seq=2 ttl=64 time=1.107 ms
84 bytes from 192.168.1.11 icmp_seq=3 ttl=64 time=2.303 ms
84 bytes from 192.168.1.11 icmp_seq=4 ttl=64 time=0.761 ms
84 bytes from 192.168.1.11 icmp_seq=5 ttl=64 time=0.934 ms

PC2-chileshe> 
```

Рис. 2: Проверка соединения между PC2 и PC1

Анализ трафика в Wireshark

- Обнаружены ARP-запросы и ответы
- Используется gratuitous ARP
- Определение соответствия IP и MAC адресов

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	::	ff02::2	ICMPv6	62	Router Solicitation
2	0.007436	::	ff02::2	ICMPv6	62	Router Solicitation
3	0.059569	Private 66:68:01	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.12 (Request)
4	0.058141	Private 66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.11 (Request)
5	1.051536	Private 66:68:01	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.12 (Request)
6	1.059449	Private 66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.11 (Request)
7	2.052985	Private 66:68:01	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.12 (Request)
8	2.059871	Private 66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.11 (Request)
.....						
▶ Frame 3: 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0						
▶ Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Broadcast (ff:ff:ff:ff:ff:ff)						
▶ Destination: Broadcast (ff:ff:ff:ff:ff:ff)						
▶ Source: Private_66:68:01 (00:50:79:66:68:01)						
Type: ARP (0x0806)						
[Stream index: 2]						
Padding: 00000000000000000000000000000000						
Frame check sequence: 0x00000000 [unverified]						
[FCS Status: Unverified]						
▶ Address Resolution Protocol (request/gratuitous ARP)						
Hardware type: Ethernet (1)						
Protocol type: IPv4 (0x0800)						
Hardware size: 6						
Protocol size: 4						
Opcode: request (1)						
[Is gratuitous: True]						
Sender MAC address: Private_66:68:01 (00:50:79:66:68:01)						
Sender IP address: 192.168.1.12						
Target MAC address: Broadcast (ff:ff:ff:ff:ff:ff)						
Target IP address: 192.168.1.12						

- Echo Request и Echo Reply
- Проверка доступности узлов
- Работа сетевого уровня

No.	Time	Source	Destination	Protocol	Length	Info
7	2.052985	Private_66:68:01	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.12 (Request)
8	2.059871	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.11 (Request)
9	140.792536	Private_66:68:00	Broadcast	ARP	64	Who has 192.168.1.12? Tell 192.168.1.11
10	140.794530	Private_66:68:01	Private_66:68:00	ARP	64	192.168.1.12 is at 00:50:79:66:68:01
11	140.797045	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) request id=0x5dd4, seq=1/256, ttl=64 (reply in 12)
12	140.797672	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) reply id=0x5dd4, seq=1/256, ttl=64 (request in 11)
13	147.639449	192.168.1.11	192.168.1.12	ECHO	98	Request
14	147.640250	192.168.1.12	192.168.1.11	ECHO	98	Response
15	151.487148	192.168.1.11	192.168.1.12	TCP	74	53277 → 7 [SYN] Seq=0 Win=2920 Len=0 MSS=1460 TSval=1765798824 TSecr=0 WS=2

Frame 12: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0	0000	00 50 79 66 68 01
Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Private_66:68:00 (00:50:79:66:68:00)	0010	00 54 d4 5d 00 00
Destination: Private_66:68:00 (00:50:79:66:68:00)	0020	01 00 00 00 00 00
Source: Private_66:68:01 (00:50:79:66:68:01)	0030	0e 0f 10 11 12 13
Type: IPv4 (0x0800)	0040	1e 1f 20 21 22 23
[Stream index: 4]	0050	2e 2f 30 31 32 33
Internet Protocol Version 4, Src: 192.168.1.12, Dst: 192.168.1.11	0060	3e 3f
0100 = Version: 4		
.... 0101 = Header Length: 20 bytes (5)		
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)		
Total Length: 84		
Identification: 0xd45d (54365)		
0000 = Flags: 0x0		
...0 0000 0000 0000 = Fragment Offset: 0		
Time to live: 64		
Protocol: ICMP (1)		
Header Checksum: 0x22e4 [validation disabled]		
[Header checksum status: Unverified]		
Source Address: 192.168.1.12		
Destination Address: 192.168.1.11		
[Stream index: 0]		
Internet Control Message Protocol		
Type: 0 (Echo (ping) reply)		
Code: 0		
Checksum: 0xca36 [correct]		
[Checksum Status: Good]		
Identifier (BE): 24020 (0x5dde)		
Identifier (LE): 54365 (0xd45d)		
Sequence Number (BE): 1 (0x0001)		
Sequence Number (LE): 256 (0x0100)		
[Request frame: 11]		
[Response time: 0.627 ms]		
Data (56 bytes)		

UDP эхо-запросы

- UDP: передача без установления соединения

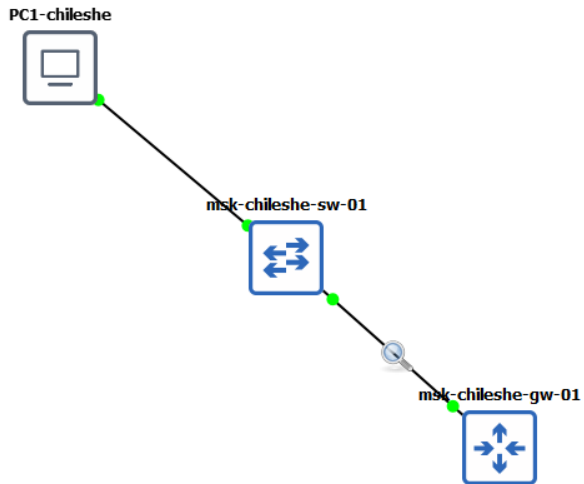
No.	Time	Source	Destination	Protocol	Length	Info
7	2.052985	Private_66:68:01	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.12 (Request)
8	2.059871	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.11 (Request)
9	140.792536	Private_66:68:00	Broadcast	ARP	64	Who has 192.168.1.12? Tell 192.168.1.11
10	140.794530	Private_66:68:01	Private_66:68:00	ARP	64	192.168.1.12 is at 00:50:79:66:68:01
11	140.797045	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) request id=0x5dd4, seq=1/256, ttl=64 (reply in 12)
12	140.797672	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) reply id=0x5dd4, seq=1/256, ttl=64 (request in 11)
13	147.630449	192.168.1.11	192.168.1.12	ECHO	98	Request
14	147.640250	192.168.1.12	192.168.1.11	ECHO	98	Response
15	151.487148	192.168.1.11	192.168.1.12	TCP	74	53277 → 7 [SYN] Seq=0 Win=2920 Len=0 MSS=1460 TSval=1765798824 TSecr=0 WS=2
Frame 13: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0						
Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Private_66:68:01 (00:50:79:66:68:01)						
Destination: Private_66:68:01 (00:50:79:66:68:01)						
Source: Private_66:68:00 (00:50:79:66:68:00)						
Type: IPv4 (0x0800)						
[Stream index: 4]						
Internet Protocol Version 4, Src: 192.168.1.11, Dst: 192.168.1.12						
0100 = Version: 4						
... 0101 = Header Length: 20 bytes (5)						
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)						
Total Length: 84						
Identification: 0xd464 (54372)						
000, = Flags: 0x0						
...0 0000 0000 0000 = Fragment Offset: 0						
Time to Live: 64						
Protocol: UDP (17)						
Header Checksum: 0x22cd [validation disabled]						
[Header checksum status: Unverified]						
Source Address: 192.168.1.11						
Destination Address: 192.168.1.12						
[Stream index: 0]						
User Datagram Protocol, Src Port: 64114, Dst Port: 7						
Source Port: 64114						
Destination Port: 7						
Length: 64						
Checksum: 0xe602 [unverified]						
[Checksum Status: Unverified]						
[Stream index: 0]						
[Stream Packet Number: 1]						
[Timestamps]						
UDP payload (56 bytes)						
Echo						
Echo data: 0050796668000e8f101112131415161718191a1b1c1d1e1f202122232425262728292a2b2c2d2e2f303132333435363738393a3b3c3d3e3f						

- ТСР: установка соединения, передача и завершение

No.	Time	Source	Destination	Protocol	Length	Info
15	151.487148	192.168.1.11	192.168.1.12	TCP	74	53277 → 7 [SYN] Seq=0 Win=2920 Len=0 MSS=1460 TSval=1765790824 TSecr=0 WS=2
16	151.487576	192.168.1.12	192.168.1.11	TCP	54	7 → 53277 [SYN, ACK] Seq=0 Ack=1 Win=2920 Len=0
17	151.488829	192.168.1.11	192.168.1.12	TCP	66	53277 → 7 [ACK] Seq=1 Ack=1 Win=2920 Len=0 TSval=1765790824 TSecr=0
18	151.489062	192.168.1.11	192.168.1.12	ECHO	122	Request
19	151.489928	192.168.1.12	192.168.1.11	TCP	54	7 → 53277 [ACK] Seq=1 Ack=57 Win=2920 Len=0
20	151.490691	192.168.1.11	192.168.1.12	TCP	66	53277 → 7 [FIN, PSH, ACK] Seq=57 Ack=1 Win=2920 Len=0 TSval=1765790824 TSecr=0
21	151.493060	192.168.1.12	192.168.1.11	TCP	54	7 → 53277 [ACK] Seq=1 Ack=58 Win=2920 Len=0
22	151.493112	192.168.1.12	192.168.1.11	TCP	54	7 → 53277 [FIN, ACK] Seq=1 Ack=58 Win=2920 Len=0
23	151.495410	192.168.1.11	192.168.1.12	TCP	66	53277 → 7 [ACK] Seq=58 Ack=2 Win=2920 Len=0 TSval=1765790824 TSecr=0

Frame 17: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface -, id 0 Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Private_66:68:01 (00:50:79:66:68:01) - Destination: Private_66:68:01 (00:50:79:66:68:01) - Source: Private_66:68:00 (00:50:79:66:68:00) - Type: IPv4 (0x0000) [Stream index: 4] Internet Protocol Version 4, Src: 192.168.1.11, Dst: 192.168.1.12 0100 = Version: 4 0101 = Header Length: 20 bytes (5) Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT) Total Length: 52 Identification: 0xd469 (54377) 0000 = Flags: 0x0 ...0 0000 0000 0000 = Fragment Offset: 0 Time to Live: 64 Protocol: TCP (6) Header Checksum: 0x22f3 [validation disabled] [Header checksum status: Unverified] Source Address: 192.168.1.11 Destination Address: 192.168.1.12 [Stream index: 0] Transmission Control Protocol, Src Port: 53277, Dst Port: 7, Seq: 1, Ack: 1, Len: 0 Source Port: 53277 Destination Port: 7 [Stream index: 0] [Stream Packet Number: 3] [Conversation completeness: Complete, WITH_DATA (31)] [TCP Segment Len: 0] Sequence Number: 1 (relative sequence number) Sequence Number (raw): 425208240 [Next Sequence Number: 1 (relative sequence number)] Acknowledgment Number: 1 (relative ack number) Acknowledgment number (raw): 1474254765 1000 = Header Length: 32 bytes (8) Flags: 0x010 (ACK) Windows: 2920 [Calculated window size: 2920] [Window size scaling factor: -2 (no window scaling used)] Checksum: 0xe78b [unverified] [Checksum Status: Unverified] Urgent Pointer: 0 Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps	0000 00 50 79 66 68 00 00 0010 00 34 d4 69 00 0020 01 0c d0 1d 00 0030 0b 68 e7 8b 00 0040 00 00
---	---

Сеть с маршрутизатором FRR

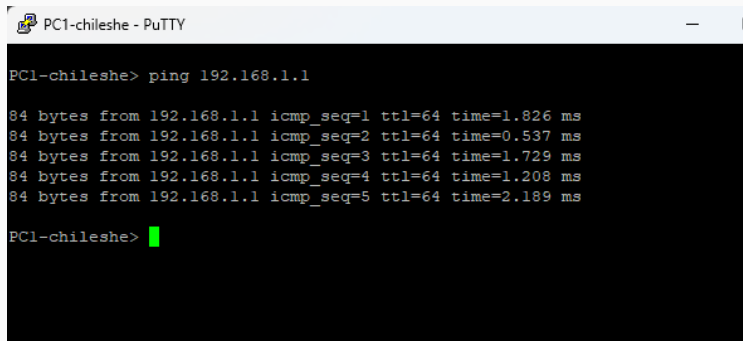


Настройка маршрутизатора FRR

```
msk-chileshe-gw-01# show running-config
Building configuration...

Current configuration:
!
frr version 8.1
frr defaults traditional
hostname frr
hostname msk-chileshe-gw-01
service integrated-vtysh-config
!
interface eth0
 ip address 192.168.1.1/24
exit
!
end
msk-chileshe-gw-01# show interface brief
Interface      Status    VRF          Addresses
-----
eth0            up        default      192.168.1.1/24
eth1            down      default
eth2            down      default
eth3            down      default
eth4            down      default
eth5            down      default
eth6            down      default
eth7            down      default
lo              up        default
pimreg          up        default

msk-chileshe-gw-01#
```



```
PC1-chileshe - PuTTY

PC1-chileshe> ping 192.168.1.1

84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=1.826 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=0.537 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=1.729 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=1.208 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=2.189 ms

PC1-chileshe> █
```

Рис. 9: Проверка связи PC1 с FRR

No.	Time	Source	Destination	Protocol	Length	Info
17	366.984763	192.168.1.1	192.168.1.11	ICMP	98	Echo (ping) reply id=0x9fd6, seq=2/512, ttl=64 (request in 16)
18	367.986666	192.168.1.11	192.168.1.1	ICMP	98	Echo (ping) request id=0xa0d6, seq=3/768, ttl=64 (reply in 19)
19	367.987840	192.168.1.1	192.168.1.11	ICMP	98	Echo (ping) reply id=0xa0d6, seq=3/768, ttl=64 (request in 18)
20	368.989396	192.168.1.1	192.168.1.1	ICMP	98	Echo (ping) request id=0xa1d6, seq=4/1024, ttl=64 (reply in 21)
21	368.990229	192.168.1.1	192.168.1.11	ICMP	98	Echo (ping) reply id=0xa1d6, seq=4/1024, ttl=64 (request in 20)
22	369.992665	192.168.1.11	192.168.1.1	ICMP	98	Echo (ping) request id=0xa2d6, seq=5/1280, ttl=64 (reply in 23)
23	369.993819	192.168.1.1	192.168.1.11	ICMP	98	Echo (ping) reply id=0xa2d6, seq=5/1280, ttl=64 (request in 22)
24	371.055343	0c:23:16:d9:00:00	Private 66:68:00	ARP	60	Who has 192.168.1.11? Tell 192.168.1.1
25	371.055699	Private 66:68:00	0c:23:16:d9:00:00	ARP	60	192.168.1.11 is at 00:50:79:66:68:00

- Frame 20: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0 - Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: 0c:23:16:d9:00:00 (0c:23:16:d9:00:00) - Internet Protocol Version 4, Src: 192.168.1.11, Dst: 192.168.1.1 0100 = Version: 4 0101 = Header Length: 20 bytes (5) - Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT) - Total Length: 84 - Identification: 0xd6a1 (54945) 0000 = Flags: 0x0 ...0 0000 0000 0000 = Fragment Offset: 0 - Time to Live: 64 - Protocol: ICMP (1) - Header Checksum: 0x20ab [validation disabled] [Header checksum status: Unverified] - Source Address: 192.168.1.11 - Destination Address: 192.168.1.1 [Stream index: 0] - Internet Control Message Protocol - Type: 8 (Echo (ping) request) - Code: 0 - Checksum: 0x7e31 [correct] [Checksum Status: Good] - Identifier (BE): 41430 (0xa1d6) - Identifier (LE): 54945 (0xd6a1) - Sequence Number (BE): 4 (0x0004) - Sequence Number (LE): 1024 (0x0400) [Response frame: 21] - Data (56 bytes)	<pre> 0000 0c 23 16 d9 0010 00 54 d6 a1 0020 01 01 08 00 0030 0e 0f 10 11 0040 1e 1f 20 21 0050 2e 2f 30 31 0060 3e 3f </pre>
--	---

Рис. 10: Анализ ICMP и ARP

Сеть с маршрутизатором VyOS

msk-chileshe-gw-01 - PuTTY

```
[edit]
vyos@vyos# delete interfaces ethernet e
eth0 eth1 eth2
[edit]
vyos@vyos# delete interfaces ethernet eth0 address dhcp
[edit]
vyos@vyos# compare
[edit interfaces ethernet eth0]
- address dhcp
+ address 192.168.1.1/24
[edit system]
> host-name msk-chileshe-gw-01
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos# show interfaces
  ethernet eth0 {
    address 192.168.1.1/24
    hw-id 0c:1e:d6:f1:00:00
  }
  ethernet eth1 {
    hw-id 0c:1e:d6:f1:00:01
  }
  ethernet eth2 {
    hw-id 0c:1e:d6:f1:00:02
  }
  loopback lo {
  }
[edit]
vyos@vyos#
```

![Проверка связи PC1 с маршрутизатором FRR](16.png){ #fig:016 width=80% }

No.	Time	Source	Destination	Protocol	Length	Info
16	102.241894	192.168.1.11	192.168.1.1	ICMP	98	Echo (ping) request id=0x9cd7, seq=1/256, ttl=64 (reply in 17)
17	102.243269	192.168.1.1	192.168.1.11	ICMP	98	Echo (ping) reply id=0x9cd7, seq=1/256, ttl=64 (request in 16)
18	103.245514	192.168.1.11	192.168.1.1	ICMP	98	Echo (ping) request id=0x9dd7, seq=2/512, ttl=64 (reply in 19)
19	103.247164	192.168.1.1	192.168.1.11	ICMP	98	Echo (ping) reply id=0x9dd7, seq=2/512, ttl=64 (request in 18)
20	104.249245	192.168.1.11	192.168.1.1	ICMP	98	Echo (ping) request id=0x9ed7, seq=3/768, ttl=64 (reply in 21)
21	104.249937	192.168.1.1	192.168.1.11	ICMP	98	Echo (ping) reply id=0x9ed7, seq=3/768, ttl=64 (request in 20)
22	105.251309	192.168.1.11	192.168.1.1	ICMP	98	Echo (ping) request id=0x9fd7, seq=4/1024, ttl=64 (reply in 23)
23	105.252482	192.168.1.1	192.168.1.11	ICMP	98	Echo (ping) reply id=0x9fd7, seq=4/1024, ttl=64 (request in 22)
24	106.255425	192.168.1.11	192.168.1.1	ICMP	98	Echo (ping) request id=0xa0d7, seq=5/1280, ttl=64 (reply in 25)

Frame 22: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0

Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: 0c:1e:d6:f1:00:00 (0c:1e:d6:f1:00:00)

Internet Protocol Version 4, Src: 192.168.1.11, Dst: 192.168.1.1

0100 = Version: 4

.... 0101 = Header Length: 20 bytes (5)

Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

Total Length: 84

Identification: 0xd79f (55199)

000. = Flags: 0x0

...0 0000 0000 0000 = Fragment Offset: 0

Time to Live: 64

Protocol: ICMP (1)

Header Checksum: 0xfad [validation disabled]

[Header checksum status: Unverified]

Source Address: 192.168.1.11

Destination Address: 192.168.1.1

[Stream index: 1]

Internet Control Message Protocol

Type: 8 (Echo (ping) request)

Code: 0

Checksum: 0x8030 [correct]

[Checksum Status: Good]

Identifier (BE): 40919 (0x9fd7)

Identifier (LE): 55199 (0xd79f)

Sequence Number (BE): 4 (0x0004)

Sequence Number (LE): 1024 (0x0400)

[Response frame: 23]

Data (56 bytes)

0000 0c 1e d6 f1 00
0010 00 54 d7 9f 00
0020 01 01 08 00 80
0030 0e 0f 10 11 11
0040 1e 1f 20 21 22
0050 2e 2f 30 31 32
0060 3e 3f

Рис. 12: Анализ ICMP и ARP-трафика для VyOS

Вывод

- Смоделированы простейшие сети в GNS3
- Настроена IP-адресация узлов и маршрутизаторов
- Подтверждена корректная связность сети
- Проанализированы протоколы ARP и ICMP
- Изучены различия работы TCP и UDP
- Практически закреплены навыки анализа трафика в Wireshark